Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Tableau Data Prep Masterclass

Course Code: C181

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 1

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	4 October 2026	Initial release of Tableau Data Prep Masterclass (C181): two-day 9:30am–5:30pm schedule.	Dr. Alfred Ang

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TABLE OF CONTENTS

Course Information **4**
Learning Outcomes **4**
Daily Schedule **4**
Day 1 — Connecting, preparing, exploring and analysing data 4
Day 2 — Dashboards, sharing, core concepts and certification readiness 5
Topic and Lab Slide Map **5**
Topic 01 — Connecting to and Preparing Data in Tableau Desktop · Slides 12–56 5
Topic 02 — Exploring and Analyzing Data in Tableau Desktop · Slides 57–96 6
Topic 03 — Sharing Insights with Dashboards and Workbooks · Slides 97–125 7
Topic 04 — Understanding Tableau Concepts and Certification Preparation · Slides 126–146 7
Resources Required **7**

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Course Information

Course Title	Tableau Data Prep Masterclass
Course Code	C181
Duration	2 days · 15 instructional hours
Daily Time	9:30 am–5:30 pm; 30-minute lunch break
Mode	Instructor-led concept teaching, demonstrations and Tableau Desktop labs
Trainer	Dr. Alfred Ang

Learning Outcomes

- LO1: Identify and connect appropriate data sources, then prepare trustworthy Tableau data models.
- LO2: Organise fields and create effective visual elements that answer business questions.
- LO3: Extract, combine and analyse data using relationships, joins, unions, filters and calculations.
- LO4: Build, format and share interactive dashboards and workbooks that communicate meaningful insights.

Daily Schedule

Day 1 — Connecting, preparing, exploring and analysing data

Time	Duration	Topic / Lab	Method	Slides
9:30–9:45	15 min	Welcome, trainer/learner introductions, outcomes and course orientation	Facilitation	Slides 1–11
9:45–10:50	**65 min**	**Topic 1 — Connecting to and Preparing Data, including Tableau Prep Builder cleansing**	**Concept + demo**	**Slides 12–56**
10:50–12:30	100 min	Labs 1–3 — connection/extract audit, relationship model, union and join	Guided hands-on	Slides 48–53
12:30–13:00	30 min	Lunch break	Lunch	—
13:00–14:10	70 min	Lab 11 — Tableau Prep cleansing flow: profile, clean, split, Group and Replace, pivot, union, join, output the flow and validate against the Control sheet	Guided hands-on	Slides 54–55
14:10–15:10	**60 min**	**Topic 2 — Exploring and**	**Concept + demo**	**Slides 57–96**

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		Analyzing Data		
15:10–17:15	125 min	Labs 4–7 — chart portfolio, organisation/filtering, calculations and geographic analysis	Guided hands-on	Slides 88–95
17:15–17:30	15 min	Day 1 exam check and Q&A	Review	Slides 57–96

Total instructional time: 450 minutes (7.5 hours), 9:30 am–5:30 pm, with a 30-minute lunch break.

Day 2 — Dashboards, sharing, core concepts and certification readiness

Time	Duration	Topic / Lab	Method	Slides
9:30–9:50	20 min	Day 1 recap	Review	Slides 57–96
9:50–11:40	**110 min**	**Topic 3 — Sharing Insights with Dashboards and Workbooks**	**Concept + demo**	**Slides 97–125**
11:40–12:30	50 min	Labs 8–9 — interactive dashboard and governed executive sharing	Guided hands-on	Slides 121–124
12:30–13:00	30 min	Lunch break	Lunch	—
13:00–14:30	90 min	Labs 8–9 continued — interactive dashboard and governed executive sharing	Guided hands-on	Slides 121–124
14:30–15:10	**40 min**	**Topic 4 concepts and certification reasoning**	**Concept + demo**	**Slides 126–146**
15:10–17:15	125 min	Lab 10 — Northstar Retail capstone and exam-domain evidence	Guided hands-on	Slides 144–145
17:15–17:30	15 min	Course wrap-up, Q&A and course feedback	Review	Slides 147–155

Total instructional time: 450 minutes (7.5 hours), 9:30 am–5:30 pm, with a 30-minute lunch break.

Topic and Lab Slide Map

Topic 01 — Connecting to and Preparing Data in Tableau Desktop · Slides 12–56

Official weighting: 23%. Connections · extracts · metadata · relationships · joins · unions · field properties · Tableau Prep cleansing

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Lab 01 — Audit a Retail Sales Connection and Extract · Slides 48–49

Goal: Connect to Excel, validate field metadata, compare live and extract behaviour. Duration: 60 minutes. Evidence: A validated Tableau data source, field-role audit, live/extract decision and refreshed .hyper extract

Lab 02 — Model Orders, Returns, Products and Targets · Slides 50–51

Goal: Create relationships and compare them with physical joins. Duration: 75 minutes. Evidence: A four-table logical model with validated cardinality, unmatched-key checks and a join-risk comparison

Lab 03 — Union Monthly Files and Join Store Attributes · Slides 52–53

Goal: Append compatible tables, merge mismatched fields and join a lookup. Duration: 60 minutes. Evidence: A three-month union with provenance fields, merged columns, a validated store join and saved data-source metadata

Lab 11 — Cleanse a Messy Sales Extract with Tableau Prep Builder · Slides 54–55

Goal: Build a repeatable Tableau Prep flow that profiles, cleans, reshapes and outputs an analysis-ready table. Duration: 75 minutes. Evidence: A saved Tableau Prep flow (.tfl) with Input, Clean, Split, Group and Replace, Pivot, Union, Join and Output steps, plus a cleaned .hyper or CSV output and a documented change list

Topic 02 — Exploring and Analyzing Data in Tableau Desktop · Slides 57–96

Official weighting: 37%. Charts · groups · sets · hierarchies · filters · analytics · calculations

Lab 04 — Build the Core Tableau Chart Portfolio · Slides 88–89

Goal: Create and justify nine chart patterns from one realistic sales dataset. Duration: 75 minutes. Evidence: Nine named worksheets: bar, line, scatterplot, symbol map, combined axis, dual axis, stacked bar, density map and highlight table

Lab 05 — Create Groups, Sets, Hierarchies and Filters · Slides 90–91

Goal: Organise customer data and apply filters at the intended analytical stage. Duration: 60 minutes. Evidence: Customer tier groups, Top-N and threshold sets, drill hierarchy, context filter and relative date controls

Lab 06 — Apply Calculations, Table Calculations and Parameters · Slides 92–93

Goal: Create reliable row-level, aggregate and view-dependent analytics. Duration: 75 minutes. Evidence: Calculated KPIs, running totals, percent-of-total, moving average, reference line and parameter-driven measure swap

Lab 07 — Analyse Southeast Asia Logistics Geographically · Slides 94–95

Goal: Create symbol, filled and density maps with trustworthy geographic context. Duration: 60 minutes. Evidence: Resolved geographic roles, three map types, tooltip context and a map-vs-bar decision note

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Topic 03 — Sharing Insights with Dashboards and Workbooks · Slides 97–125

Official weighting: 25%. Formatting · dashboards · actions · device layouts · stories · export · publish

Lab 08 — Build an Interactive Service Operations Dashboard · Slides 121–122

Goal: Compose worksheets, containers, filters, tooltips and dashboard actions. Duration: 90 minutes. Evidence: A 1366×768 operational dashboard with KPIs, four views, scoped filters, highlight/filter actions and accessible tooltips

Lab 09 — Optimise, Package and Share an Executive Workbook · Slides 123–124

Goal: Create device layouts, a story, exports and a governed sharing plan. Duration: 60 minutes. Evidence: Desktop and phone layouts, three story points, TWB/TWBX comparison, PDF/PNG/PPT exports and a publishing checklist

Topic 04 — Understanding Tableau Concepts and Certification Preparation · Slides 126–146

Official weighting: 15%. Dimensions · measures · discrete · continuous · aggregation · exam strategy

Lab 10 — Capstone: Northstar Retail Executive Workbook · Slides 144–145

Goal: Integrate all four exam domains and complete certification-readiness evidence. Duration: 100 minutes. Evidence: A validated multi-table model, eight analytical worksheets, executive dashboard, phone layout, story, packaged workbook and exam-domain checklist

Resources Required

- Tableau Desktop (recommended version matching the certification guide where available)
- Tableau Prep Builder (for the Topic 1 data-cleansing lab)
- Microsoft Excel or compatible viewer
- Eleven individual lab folders with Excel workbooks, starter.twb and acceptance checklists
- Learner Guide and course slides
- Access to the course GitHub repository for the lab folders

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